pre-test questions the basis of questions to be presented per topic in term 2: ---

TERM 2 – COMPLETE QUESTION SET

TOPIC 1 · ECOSYSTEM'S CARRYING CAPACITY AND POPULATION GROWTH (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

The maximum number of individuals an ecosystem can support long-term without exhausting its resources is called:
A. Population density
B. Carrying capacity ☑
C. Birth rate
D. Death rate
Any factor that prevents a population from growing larger is called a:
A. Limiting factor ☑
B. Density factor
C. Growth factor
D. Carrying factor
Which is NOT a common limiting factor for most animal populations?
A. Clean water
B. Shelter or space
C. Body color of individuals ☑
D. Adequate food
When a population grows rapidly with no limiting factors, it forms a:
A. Flat line
B. S-shaped curve
C. Wavy line
D. J-shaped curve (exponential growth) ☑
As a population approaches its carrying capacity, its growth rate:
A. Slows down and approaches zero ☑
B. Speeds up
C. Becomes infinite
D. Stays exactly the same
The S-shaped population growth pattern is called:
A. Boom-and-bust growth
B. Exponential growth
C. Logistic growth ☑
D. Negative growth
A lack of available mates in a small population will:
A. Cause exponential growth
B. Limit population growth ☑
C. Increase carrying capacity
D. Have no effect
In a logistic growth curve, the flat portion at the top represents:
A. Rapid growth
B. Population crash
C. Exponential phase
D. Carrying capacity (stable population size) ☑
A drought causes most plants in a grassland to die, leading to a decline in rabbit numbers. The drought is a:
A. Limiting factor ☑
B. Growth factor
C. Carrying capacity increase
D. Mate factor
Population density refers to:
A. Birth rate minus death rate
B. Carrying capacity
C. Number of individuals per unit area or volume ☑
D. Total number of individuals only
A forest originally supports 500 deer. After a typhoon destroys much of the habitat, the carrying capacity will most likely:
A. Double to 1000

B. Stay the same
C. Decrease ☑
D. Become unlimited
Which is an example of a density-dependent limiting factor?
A. Disease or food shortage ☑
B. Typhoon
C. Earthquake
D. Volcanic eruption
Which is a density-independent limiting factor?
A. Predation
B. Competition for mates
C. Parasitism
D. Super typhoon ☑
Why does population growth slow as it nears carrying capacity?
A. Carrying capacity keeps increasing
B. Limiting factors disappear
C. All organisms stop reproducing
D. Competition for resources becomes stronger ☑
Rats increase in a rice field, followed by an increase in snakes that feed on them. Snake predation is a:
A. Biotic limiting factor ☑
B. Carrying capacity increaser
C. Abiotic factor
D. Growth promoter
In nature, what usually happens after exponential growth?
A. It continues forever
B. Limiting factors slow growth or cause a population crash ☑
C. Carrying capacity disappears
D. Resources become unlimited
If a population is below carrying capacity, it will usually:
A. Go extinct
B. Stay exactly the same
C. Increase toward carrying capacity ☑
D. Decrease rapidly
A landfill destroys most of a frog's breeding pond. Which limiting factor is affected the most?
A. Water
B. Food
C. Mates
D. Shelter or breeding space ☑
Which pair consists of abiotic limiting factors?
A. Water availability and temperature extremes ☑
B. Predators and parasites
C. Competition and disease
D. Food supply and mates
The current growth of the human population most closely follows which pattern?
A. Stable S-curve
B. Rising J-curve or early logistic growth ☑
C. Rapid decline
D. Zero growth
What is the correct sequence of logistic growth?
A. Stable → Fast → Slow
B. Crash → Fast → Stable
C. Slow → Fast → Slowing → Stable at carrying capacity ☑
D. Stable → Crash → Fast
If a population exceeds its carrying capacity, what is most likely to happen?
A. Growth continues rapidly
B. Carrying capacity permanently increases
C. A population crash occurs because of resource shortages ☑
D. Nothing changes
A lake can support 1,000 tilapia, but 1,200 are introduced. What will most likely happen?
A. The carrying capacity becomes 1,200
B. All survive
C. Exponential growth continues

D. Many die until the population returns near carrying capacity ☑
Why are mates considered a limiting factor in small isolated populations?
A. Individuals have difficulty finding mates, reducing reproduction ☑
B. Mates increase carrying capacity
C. Food competition increases
D. Males always fight
What is the main difference between density-dependent and density-independent limiting factors?
A. Independent factors are always living things
B. Density-dependent effects become stronger as population density increases ☑
C. Density-dependent factors never cause extinction
D. They have exactly the same effects
Which situation best illustrates a population slowing as it approaches carrying capacity?
A. A volcano kills most birds
B. A new species grows without limits
C. Mouse population: 100 → 180 → 210 → 215 → 212 ☑
D. Disease wipes out rabbits
How does competition affect populations near carrying capacity?
A. It increases resources
B. It removes predators
C. It automatically raises carrying capacity
D. Fewer individuals survive and reproduce because resources are limited ☑
Protecting both food sources and nesting sites for the Philippine Eagle helps because:
A. Carrying capacity depends on all essential resources ☑
B. Eagles do not need mates
C. Only food matters
D. Carrying capacity never changes
Why can't exponential population growth continue forever?
A. Death rates always decrease
B. Resources are limited in every ecosystem ☑
C. Carrying capacity increases endlessly
D. Birth rates suddenly stop
A population stabilizes at carrying capacity when:
A. Death rate becomes zero
B. No limiting factors exist
C. Birth rate is approximately equal to death rate ☑

1. D. Birth rate becomes zero

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TOPIC 2 · BIOTECHNOLOGY (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

Biotechnology is best defined as:
A. Only genetic engineering
B. Growing plants only
C. Using living organisms or biological processes to make useful products ☑
D. Only laboratory work
Traditional food biotechnology mainly depends on:
A. Fermentation by microorganisms ☑
B. Freezing
C. Canning
D. Boiling
Nata de coco is produced by fermenting coconut water with:
A. Mold
B. Yeast
C. Acetobacter xylinum ☑
D. Lactic acid bacteria
Vinegar is produced when alcohol is converted into:
A. Citric acid
B. Acetic acid ☑
C. Hydrochloric acid

D. Lactic acid
Cheese, yogurt, and pickles are made using bacteria that produce:
A. Lactic acid ☑
B. Oxygen
C. Alcohol
D. Acetic acid
Soy sauce is traditionally made from:
A. Coconut milk
B. Rice only
C. Fruit juice
D. Soybeans, grains, mold, and bacteria ☑
GMOs are organisms whose ____ has been altered in a laboratory.
A. Cell size
B. DNA or genetic material ☑
C. Food value only
D. Color only
In Vitro Fertilization (IVF) takes place:
A. Outside the body before implantation ☑
B. Without sperm and egg
C. Inside the body
D. With one parent only
One benefit of pest-resistant GMO crops is:
A. They never spoil
B. Need more water
C. Less pesticide use and higher yield ☑
D. Better taste
Which is an ethical concern related to IVF?
A. It always causes disease
B. It always succeeds
C. It produces only identical twins
D. Issues involving embryos, cost, and fairness ☑
Which list contains only traditional fermented foods?
A. Cheese, soy sauce, vinegar, nata de coco, and yogurt ☑
B. Bt corn
C. Modern medicines
D. Insulin and vaccines
Baker's yeast helps bread rise by producing:
A. Oxygen
B. Acetic acid
C. Carbon dioxide and alcohol ☑
D. Lactic acid
Golden Rice was developed to provide more:
A. Iron
B. Beta-carotene (Vitamin A precursor) ☑
C. Protein
D. Vitamin C
One environmental concern about Bt corn is:
A. Pests may become resistant and non-target insects may be affected ☑
B. It stops photosynthesis
C. It requires much more water
D. It kills all nearby plants
One concern about GMOs for Filipino farmers is:
A. They are always poisonous
B. They grow only in cold climates
C. Patented seeds may increase farming costs ☑
D. They always taste worse
Most modern insulin is produced by:
A. Blood donations
B. Bacteria carrying the human insulin gene ☑
C. Chemical synthesis only
D. Pig pancreas only
Which pairing is correct?

A. Vinegar – Acetic acid bacteria ☑
B. Bread – Mold
C. Yogurt – Yeast only
D. Nata de coco – Lactobacillus
Fermentation is considered ancient biotechnology because:
A. It requires modern laboratories
B. It uses DNA editing
C. It is only for modern foods
D. It was practiced long before microbes were understood ☑
Cloning produces offspring that are:
A. Genetically identical to one parent ☑
B. Always stronger
C. Unable to reproduce
D. Genetically unique
A major ethical issue in gene editing with CRISPR is:
A. It only works on plants
B. Unintended effects and designer baby concerns ☑
C. It cures nothing
D. It is completely safe
What is the correct sequence in traditional vinegar production?
A. Alcohol → Sugar
B. Lactic acid → Alcohol
C. Sugar → Yeast → Alcohol → Bacteria → Acetic acid ☑
D. Sugar → Acid directly
Nata de coco is formed from:
A. Milk protein
B. Yeast foam
C. Bacterial cellulose ☑
D. Boiled coconut water
The Cartagena Protocol focuses on:
A. Cheese making
B. Human cloning
C. IVF regulations
D. Safe handling and trade of genetically modified organisms ☑
An environmental benefit of approved GMO crops is:
A. Reduced use of chemical pesticides and fertilizer runoff ☑
B. They replace all wild plants
C. They grow only in one season
D. They require more fertilizer
One economic concern about IVF is:
A. It always causes infertility
B. High cost limits access for many families ☑
C. It always produces boys
D. It is always dangerous
Fermentation helps preserve food because it:
A. Dries food completely
B. Uses only salt
C. Produces acids or alcohol that slow spoilage microorganisms ☑
D. Heats food
Bt corn contains a gene that:
A. Makes the plant glow
B. Eliminates the need for water
C. Makes corn sweeter
D. Produces a protein harmful to certain insect pests ☑
Traditional and modern biotechnology differ because:
A. Traditional uses selection and fermentation, while modern changes DNA directly ☑
B. Traditional never uses microbes
C. Modern is always safer
D. Only modern biotechnology produces food
One ethical trade-off in IVF is:
A. Success is guaranteed
B. There is no cost involved

C. Extra embryos may be frozen or discarded ☑
D. There are no risks
Compared with traditional breeding, modern biotechnology:
A. Is only used in rich countries
B. Changes specific genes directly ☑
C. Cannot improve crop yield
D. Uses no science

TOPIC 3 · PLATE TECTONICS(only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. **Theory that Earth's outer shell moves in slabs:**
   A. Earthquake theory
   B. Plate Tectonics
   C. Continental drift
   D. Seafloor spreading
   ☑ **B** — Plate tectonics unifies how plates move and interact.
2. **The rigid moving slabs are called:**
   A. Fault blocks
   B. Asthenosphere
   C. Mantle plumes
   D. Tectonic / lithospheric plates
   ☑ **D** — The lithosphere is broken into moving plates.
3. **Plates move at about the speed of:**
   A. A runner
   B. Fingernails growing
   C. A car
   D. A jet
   ☑ **B** — Typical speed is a few centimeters per year.
4. **Modern tool to measure plate motion precisely:**
   A. Thermometer
   B. GPS satellites
   C. Compass
   D. Barometer
   ☑ **B** — GPS tracks position to millimeters over time.
5. **Plates moving apart:**
   A. Subduction
   B. Transform
   C. Divergent
   D. Convergent
   ☑ **C** — Divergent boundaries create new crust.
6. **Plates colliding:**
   A. Divergent
   B. Ridge
   C. Transform
   D. Convergent
   ☑ **D** — Convergent boundaries destroy or thicken crust.
7. **Plates sliding past each other:**
   A. Trench
   B. Convergent
   C. Transform fault
   D. Divergent
   ☑ **C** — Transform boundaries have no creation or destruction of crust.
8. **Soft flowing layer under plates:**
   A. Crust
   B. Core
   C. Asthenosphere
   D. Lithosphere
   ☑ **C** — The asthenosphere allows plates to move.
9. **One plate sinking under another:**
   A. Spreading
   B. Faulting

C. Subduction
D. Rifting
☑ **C** — Dense oceanic plate sinks into the mantle.

10. **Main fault system in the Philippines:**
A. East African Rift
B. San Andreas
C. Philippine Fault System
D. Marikina Fault only
☑ **C** — It runs the length of the archipelago.
11. **Continent-continent collision forms:**
A. Tall non-volcanic mountains
B. Island arcs
C. Trenches
D. Volcanoes
☑ **A** — No subduction means no melting for magma.
12. **Ocean-continent collision produces:**
A. Mid-ocean ridge
B. Trench + continental volcanic arc
C. Rift valley
D. Transform fault
☑ **B** — Subduction melts rock to feed volcanoes on land.
13. **The Philippines sits between:**
A. Nazca and South American Plates
B. Indian and African Plates
C. Philippine Sea Plate and Eurasian Plate
D. Pacific and North American Plates
☑ **C** — Interaction of these plates forms our geology.
14. **Many Philippine volcanoes and trenches exist because:**
A. No plate motion
B. Several plates are subducting around us
C. We are on a hot spot only
D. Plates are moving apart
☑ **B** — Multiple subduction zones ring the country.
15. **Transform boundaries commonly have:**
A. Strong shallow earthquakes
B. Trenches
C. Volcanoes
D. New crust
☑ **A** — Friction builds then slips in quakes; no melting.
16. **Himalayas formed when:**
A. Indian Plate collided with Eurasian Plate
B. Plates moved apart
C. Ocean plate melted
D. Plates slid sideways
☑ **A** — Two continents crumpled together.
17. **Andes Mountains formed from:**
A. Rifting
B. Continent-continent push
C. Hot spot
D. Nazca Plate subducting under South America
☑ **D** — Ocean plate diving under the continent.
18. **Main driver of plate motion:**
A. Wind
B. Earth's spin
C. Mantle convection currents
D. Moon pull
☑ **C** — Heat rising and sinking moves the plates.
19. **Subduction leads to:**
A. No change
B. Melting, magma, volcanoes, folded mountains
C. Only erosion
D. Rifting
☑ **B** — Sinking rock heats, melts, and rises.

20. **New seafloor is created at:**
    A. Faults
    B. Volcanoes
    C. Trenches
    D. Mid-ocean ridges / divergent boundaries
    ☑ **D** — Magma rises, cools, and adds new crust.
21. **Plate moving 7 cm/year for 50 million years travels:**
    A. 700 km
    B. 350 km
    C. 3500 km
    D. 7000 km
    ☑ **C** — 7 cm × 50,000,000 = 3,500 km.
22. **Why is GPS better than older methods?**
    A. Precise, continuous, global
    B. Works only on land
    C. Needs no power
    D. Cheaper only
    ☑ **A** — It gives accurate real-time motion data.
23. **Mariana and Philippine Trenches are:**
    A. Hot spots
    B. Rift valleys
    C. Transform features
    D. Subduction trenches
    ☑ **D** — They mark where plates sink.
24. **Correct match for the Philippines:**
    A. Divergent → Mayon
    B. Transform → Taal
    C. Convergent → Mayon Volcano + Manila Trench
    D. Hot spot → Manila Trench
    ☑ **C** — Both come from subduction at convergent boundaries.
25. **No volcanoes at the Himalayas because:**
    A. No dense ocean plate to sink and melt
    B. No earthquakes
    C. Too dry
    D. Too cold
    ☑ **A** — Continental rock is too light to subduct.
26. **Asthenosphere is:**
    A. Rigid metal
    B. Cold brittle rock
    C. Hot solid rock that flows slowly
    D. Liquid rock
    ☑ **C** — It is solid but ductile over millions of years.
27. **Trenches on both sides of the Philippines mean:**
    A. No plate motion
    B. Plates are pulling apart
    C. We are squeezed between subduction zones
    D. Only earthquakes
    ☑ **C** — The archipelago sits in a collision zone.
28. **Philippine Fault System is a:**
    A. Divergent rift
    B. Left-lateral transform fault
    C. Hot spot
    D. Subduction zone
    ☑ **B** — Blocks slide past each other horizontally.
29. **Ocean-ocean subduction makes ______; ocean-continent makes ______:**
    A. Mountains / plains
    B. No trench / no trench
    C. Volcanic island arc / continental volcanic arc
    D. Rift / ridge
    ☑ **C** — One makes islands, the other land mountains.
30. **Origin of the Philippines:**
    A. Ocean floor uplifted all at once
    B. Meteor impact

C. Merged small volcanic island arcs
D. One continent split
☑ **C** — Pieces of crust collided and stuck together.

TOPIC 4 · GLOBAL CLIMATE (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. Climate is the long-term average of:
   A. Weather
   B. Climate
   C. Season
   D. Forecast
   ☑ **A** — Climate describes average weather conditions over many years.
2. Which term refers to the increase in Earth's average temperature?
   A. Ice age
   B. Coriolis effect
   C. Global warming
   D. Cooling
   ☑ **C** — It is the long-term rise in global air and ocean temperatures.
3. Climate change includes:
   A. Only hotter summers
   B. Changes in rainfall, sea level, and temperature
   C. Daily weather changes only
   D. Air pollution only
   ☑ **B** — Climate change involves many long-term environmental changes.
4. Gases that trap heat in Earth's atmosphere are called:
   A. Noble gases
   B. Nitrogen gases
   C. Ozone gases
   D. Greenhouse gases
   ☑ **D** — These gases help keep heat near Earth's surface.
5. The main greenhouse gas released by human activities is:
   A. Nitrogen
   B. Argon
   C. Carbon dioxide (CO_2)
   D. Ozone
   ☑ **C** — Burning fossil fuels releases large amounts of CO_2.
6. The natural greenhouse effect is important because it:
   A. Makes Earth too hot
   B. Keeps Earth warm enough for life
   C. Causes storms
   D. Freezes the planet
   ☑ **B** — Without it, Earth would be too cold to support life.
7. The enhanced greenhouse effect happens when:
   A. Less heat is trapped
   B. Extra greenhouse gases trap more heat
   C. Heat escapes faster
   D. Earth's atmosphere disappears
   ☑ **B** — Human activities strengthen the natural greenhouse effect.
8. Melting glaciers and ice sheets directly cause:
   A. Lower sea level
   B. Less rainfall
   C. Sea level rise
   D. Cooler oceans
   ☑ **C** — Melted land ice adds more water to the oceans.
9. Ice core records show that modern CO_2 levels are:
   A. Lower than before
   B. About the same as always
   C. Slightly lower than natural levels
   D. Higher than at any time in the last 800,000 years
   ☑ **D** — Today's CO_2 levels are far above natural historical levels.

10. Besides melting ice, sea level rises because:
    A. Oceans become saltier
    B. Warm water expands
    C. More rain falls into the ocean
    D. Winds push water upward
    ☑ **B** — Heated water occupies more space.
11. Which group contains major greenhouse gases?
    A. Oxygen, nitrogen, helium
    B. CO_2, methane, nitrous oxide, CFCs, water vapor
    C. Hydrogen, helium, argon
    D. Dust and smoke
    ☑ **B** — These gases absorb and re-emit heat.
12. Which human activities produce large amounts of methane?
    A. Volcanoes only
    B. Cars only
    C. Cattle, rice fields, landfills, and gas leaks
    D. Oceans only
    ☑ **C** — Agriculture and waste are major methane sources.
13. CFCs are harmful because they:
    A. Cool the atmosphere
    B. Trap heat and damage the ozone layer
    C. Occur naturally in large amounts
    D. Remove greenhouse gases
    ☑ **B** — They are powerful greenhouse gases and ozone-depleting chemicals.
14. Which is strong evidence of climate change?
    A. One unusually hot day
    B. Rising global temperatures and shrinking glaciers
    C. A rainy afternoon
    D. One powerful typhoon
    ☑ **B** — Multiple long-term observations confirm climate change.
15. Deforestation increases global warming because:
    A. Trees release more oxygen
    B. Forests release stored carbon and absorb less CO_2
    C. It creates more rainfall
    D. It lowers Earth's temperature
    ☑ **B** — Trees store carbon and remove CO_2 from the atmosphere.
16. Which is NOT considered evidence of climate change?
    A. One hot day
    B. Retreating glaciers
    C. Rising sea level
    D. Earlier flowering of plants
    ☑ **A** — Single weather events do not represent climate.
17. Which sequence correctly explains human-caused global warming?
    A. More trees → More warming
    B. Burning fossil fuels → More CO_2 → Stronger greenhouse effect → Warming
    C. Less sunlight → Warming
    D. More clouds → Cooling
    ☑ **B** — Human emissions increase greenhouse gases, causing warming.
18. Why will Earth continue warming even if emissions stop immediately?
    A. Volcanoes become stronger
    B. CO_2 remains in the atmosphere for centuries and oceans store heat
    C. The Sun keeps getting hotter
    D. Winds trap heat permanently
    ☑ **B** — Past emissions continue affecting the climate.
19. The largest human source of CO_2 emissions is:
    A. Volcanoes
    B. Burning coal, oil, and natural gas
    C. Oceans
    D. Animals
    ☑ **B** — Fossil fuel combustion is the biggest contributor.
20. Scientists know extra atmospheric CO_2 comes mainly from fossil fuels because:
    A. Carbon isotope evidence matches fossil fuels
    B. Volcanoes release the same amount

C. The atmosphere is heavier
D. Ocean temperatures prove it
☑ **A** — Carbon isotope signatures identify fossil fuel emissions.

21. The difference between the natural and enhanced greenhouse effect is that:
A. Both are caused by humans
B. Natural warming supports life, while enhanced warming is caused by extra greenhouse gases
C. Natural warming is harmful
D. Enhanced warming cools Earth
☑ **B** — Human activities increase the natural greenhouse effect.
22. Arctic ice-albedo feedback means:
A. More ice absorbs more heat
B. Less ice exposes darker water, causing more warming and melting
C. Ice cools the atmosphere permanently
D. There is no effect on temperature
☑ **B** — This positive feedback speeds up Arctic warming.
23. Even a 1°C increase in global temperature is important because it:
A. Causes almost no changes
B. Increases heatwaves, floods, and stronger storms
C. Only affects polar regions
D. Makes oceans colder
☑ **B** — Small global averages can produce major regional impacts.
24. Which chain best explains stronger typhoons in the Philippines?
A. More CO_2 → Less rainfall
B. More greenhouse gases → Warmer oceans → Stronger and wetter typhoons
C. Hotter weather → Fewer storms
D. Cooler oceans → Stronger storms
☑ **B** — Warm ocean water provides more energy for tropical cyclones.
25. Why is the Arctic warming faster than most other places?
A. It receives more sunlight
B. Ice-albedo feedback accelerates warming
C. Stronger winds increase temperature
D. More people live there
☑ **B** — Melting ice exposes darker surfaces that absorb more heat.
26. Why is the statement "Climate changed naturally before, so today's change is natural" incorrect?
A. Climate has never changed naturally
B. Natural factors cannot explain today's rapid warming; human emissions do
C. Earth is always warming
D. Ice ages never happened
☑ **B** — The speed and cause of current warming match human activities.
27. Ocean acidification happens because:
A. Overfishing lowers pH
B. CO_2 dissolves in seawater, forming carbonic acid
C. Warm water becomes acidic by itself
D. Pollution alone causes acidity
☑ **B** — Carbonic acid lowers ocean pH and harms marine life.
28. Which is a major climate impact already observed in the Philippines?
A. Colder temperatures every year
B. Rising sea levels, stronger typhoons, longer dry seasons, and heavier rains
C. Less rainfall everywhere
D. No noticeable changes
☑ **B** — These are documented climate-related impacts.
29. Why isn't water vapor considered the main cause of modern climate change?
A. It does not trap heat
B. It increases after warming begins, acting as a feedback rather than the original cause
C. It is disappearing from the atmosphere
D. It only exists inside clouds
☑ **B** — CO_2 triggers warming; water vapor amplifies it.
30. Ice core records show that atmospheric CO_2 and global temperature:
A. Have no relationship
B. Closely rise and fall together, with today's CO_2 much higher than past levels
C. Always move in opposite directions
D. Change randomly
☑ **B** — Ice core evidence strongly supports the greenhouse effect.

TOPIC 5 · GLOBAL INTERACTIONS (ENSO) (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

ENSO stands for:
A. Weather System
B. El Niño Southern Oscillation ☑
C. Ocean Current
D. Wind Pattern
El Niño refers to:
A. Unusually warm eastern and central Pacific Ocean ☑
B. No wind
C. Cool Pacific waters
D. Rain everywhere
La Niña is characterized by:
A. No change
B. Global drought
C. Unusually cool eastern Pacific Ocean ☑
D. Warm Pacific waters
ENSO develops mainly in the:
A. Atlantic Ocean
B. Indian Ocean
C. Arctic Ocean
D. Tropical Pacific Ocean ☑
The normal winds that blow from east to west across the Pacific are called:
A. Trade winds ☑
B. Monsoons
C. Westerlies
D. Jet streams
El Niño usually brings which condition to the Philippines?
A. More typhoons
B. Hotter weather and less rainfall ☑
C. Cooler temperatures
D. Flooding
La Niña commonly causes:
A. No storms
B. Drought
C. Above-normal rainfall and flooding ☑
D. Hotter weather
The "Southern Oscillation" refers to the:
A. Ocean temperature
B. Sea level
C. Ocean waves
D. Air pressure changes across the Pacific ☑
Most ENSO events last about:
A. 9–12 months, sometimes up to 2 years ☑
B. One week
C. Forever
D. Ten years
PAGASA declares an ENSO event when:
A. Farmers observe unusual weather
B. One hot day occurs
C. Ocean and atmospheric conditions meet standards for several months ☑
D. News agencies announce it
During El Niño, the trade winds:
A. Become stronger
B. Weaken or sometimes reverse direction ☑
C. Stay the same
D. Disappear permanently
During El Niño, the warm pool of ocean water:
A. Moves north
B. Sinks
C. Shifts eastward across the Pacific ☑
D. Stays near Asia

The Philippines usually experiences fewer typhoons during El Niño because:
A. Storms form farther east and stronger wind shear weakens them ☑
B. Winds completely stop
C. The ocean becomes too cold
D. There is no evaporation
During La Niña, more typhoons affect the Philippines because:
A. Less moisture is available
B. Ocean temperatures decrease
C. There is less heat
D. Warm water remains near the Philippines with weaker wind shear ☑
Which sector is most affected by El Niño?
A. Rain-fed agriculture such as rice and corn farming ☑
B. Manufacturing
C. Factories
D. Urban businesses
Which combination is commonly associated with El Niño?
A. More rainfall
B. Drought, crop losses, and possible power shortages ☑
C. Cooler temperatures
D. Flooding everywhere
La Niña often results in:
A. Water shortage
B. Less wind
C. Heavy rainfall, floods, and landslides ☑
D. Drought
ENSO affects countries differently because:
A. It is completely random
B. It affects only the Pacific Ocean
C. It changes global wind and rainfall patterns ☑
D. There is no scientific reason
Why is monitoring ENSO important?
A. It helps communities prepare for possible disasters ☑
B. It changes the weather
C. It has no practical use
D. It is only for scientists
Neutral ENSO conditions mean:
A. Always dry weather
B. Climate conditions remain close to average ☑
C. Always wet weather
D. No weather occurs
Walker Circulation is:
A. A large east-west circulation of air across the Pacific ☑
B. An ocean current
C. A wave pattern
D. A type of wind
Why does El Niño often cause floods in the Americas but drought in the Philippines?
A. Mountains block rainfall
B. The Philippines is too far away
C. Rainfall shifts eastward with the warm ocean water ☑
D. It happens randomly
Climate change may affect ENSO by making:
A. Only La Niña occur
B. Extreme ENSO events more frequent or intense ☑
C. ENSO disappear
D. ENSO weaker
Which sequence correctly describes El Niño?
A. Cooler water → more storms
B. No change occurs
C. Weak trade winds → warm water shifts east → rainfall moves away → drought ☑
D. Stronger winds → more rainfall
Metro Manila may experience water shortages during El Niño because:
A. Reservoirs receive less rainfall ☑

B. There are no rivers
C. People always waste water
D. Pipe leaks are the only cause
Which statement correctly describes typhoon tracks?
A. Typhoons always follow the same path
B. El Niño often shifts storms away, while La Niña increases Philippine landfalls ☑
C. ENSO has no effect
D. Storms always hit the Visayas
The 2015–2016 Super El Niño caused:
A. A cooler-than-normal year
B. Better harvests
C. Severe drought, extreme heat, and major crop losses ☑
D. Widespread flooding
ENSO affects the Philippine monsoon because:
A. El Niño weakens Habagat, while La Niña strengthens it ☑
B. Monsoons stop completely
C. Only Amihan changes
D. There is no effect
Why do ENSO impacts differ from one event to another?
A. PAGASA changes its rules
B. Only temperature matters
C. Event strength, timing, location, and climate change all influence the effects ☑
D. It is completely random
Why is it important for students to understand ENSO?
A. It is only needed for examinations
B. It influences food, water, electricity, and public safety ☑
C. It only affects farmers
D. It has no effect on daily life
Compose
Write to Sean Alzate

TOPIC 6 · GLOBAL AND LOCAL SUSTAINABILITY (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. Actions that reduce or prevent greenhouse gas emissions are called:
   A. Conservation
   B. Mitigation
   C. Restoration
   D. Adaptation
   ☑ **B** — Mitigation focuses on reducing the causes of climate change.
2. The process of adjusting to the effects of climate change is known as:
   A. Prevention
   B. Adaptation
   C. Reduction
   D. Mitigation
   ☑ **B** — Adaptation helps people and ecosystems cope with climate impacts.
3. Which renewable energy source uses heat from beneath Earth's surface?
   A. Solar
   B. Geothermal
   C. Wind
   D. Hydropower
   ☑ **B** — Geothermal energy comes from underground heat.
4. Electricity generated from moving water is called:
   A. Biomass
   B. Hydropower
   C. Solar energy
   D. Geothermal
   ☑ **B** — Rivers and dams generate hydropower.
5. Which renewable energy source converts sunlight directly into electricity?
   A. Coal
   B. Solar energy
   C. Wind energy

D. Hydropower
☑ **B** — Solar panels capture energy from sunlight.

6. The largest source of greenhouse gas emissions in the Philippines comes from:
A. Forests
B. Burning coal, oil, and natural gas for electricity and transportation
C. Agriculture
D. Waste disposal
☑ **B** — Fossil fuels are the country's primary source of emissions.
7. Which is the BEST way for individuals to reduce transportation emissions?
A. Use private vehicles more often
B. Walk, bike, or use public transportation
C. Travel by airplane frequently
D. Buy more plastic products
☑ **B** — Reducing private vehicle use lowers emissions.
8. What do the "3 Rs" stand for?
A. Repair, Replace, Remove
B. Read, Write, Repeat
C. Reduce, Reuse, Recycle
D. Run, Rest, Recover
☑ **C** — Reducing waste comes first, followed by reusing and recycling.
9. Trees help reduce climate change by:
A. Blocking sunlight only
B. Absorbing and storing carbon dioxide (carbon sequestration)
C. Producing oxygen only
D. Cooling the air without storing carbon
☑ **B** — Forests remove CO_2 from the atmosphere and store it.
10. Which is a common impact of climate change in the Philippines?
A. Colder oceans
B. Stronger typhoons, rising sea levels, longer droughts, and hotter days
C. More fish populations
D. Longer harvest seasons
☑ **B** — These impacts are already being observed.
11. Which of the following is a renewable energy resource found in the Philippines?
A. Coal and oil only
B. Geothermal, hydro, solar, wind, biomass, and ocean energy
C. Coal only
D. Oil only
☑ **B** — The Philippines has abundant renewable energy resources.
12. Renewable energy helps reduce climate change because it:
A. Produces little or no greenhouse gases
B. Is cheaper in every situation
C. Uses more fuel efficiently
D. Makes electricity faster
☑ **A** — Renewable energy emits very little CO_2 during operation.
13. Rising sea levels can cause coastal communities to experience:
A. More beaches
B. Saltwater intrusion, erosion, flooding, and stronger storm surges
C. Cooler seawater
D. No significant effects
☑ **B** — Higher sea levels worsen coastal hazards.
14. Saving electricity helps fight climate change because:
A. It reduces the amount of fuel burned for power generation
B. It only lowers electricity bills
C. It reduces the need for electrical wires
D. It has no effect on emissions
☑ **A** — Lower electricity demand means fewer fossil fuels are burned.
15. Which household action has the greatest impact on reducing carbon emissions?
A. Recycling alone
B. Using energy-efficient appliances, conserving electricity, and installing solar panels
C. Buying less plastic only
D. Driving less once a month
☑ **B** — Reducing home energy use significantly lowers emissions.
16. Composting organic waste helps reduce climate change because it:
A. Makes soil darker
B. Prevents methane production in landfills and reduces the need for synthetic fertilizers
C. Saves water only
D. Produces more oxygen
☑ **B** — Composting lowers methane emissions from waste.

17. Republic Act No. 9513, also known as the Renewable Energy Act, aims to:
    A. Ban renewable energy development
    B. Promote renewable energy and reduce dependence on imported fossil fuels
    C. Increase coal consumption
    D. Eliminate hydropower projects
    ☑ **B** — The law supports clean and sustainable energy.
18. Climate change affects agriculture by causing:
    A. More predictable seasons
    B. Unpredictable weather that reduces crop yields and affects food prices
    C. Faster crop growth everywhere
    D. Fewer agricultural pests
    ☑ **B** — Changing weather patterns make farming more difficult.
19. Mangrove forests are important because they:
    A. Beautify coastlines only
    B. Store blue carbon and protect coastal communities from storms
    C. Block fishing boats
    D. Slow waves only
    ☑ **B** — Mangroves provide both climate mitigation and adaptation benefits.
20. The National Renewable Energy Program (NREP) aims to achieve renewable energy shares of:
    A. 100% immediately
    B. 35% by 2030 and 50% by 2040
    C. 0% renewable energy
    D. No renewable energy targets
    ☑ **B** — These are the Philippine government's renewable energy goals.
21. Which statement correctly distinguishes mitigation from adaptation?
    A. They mean the same thing
    B. Mitigation reduces emissions, while adaptation adjusts to climate impacts
    C. Both are only government responsibilities
    D. Adaptation prevents greenhouse gas emissions
    ☑ **B** — One addresses the cause; the other manages the effects.
22. Geothermal energy is especially important in the Philippines because it:
    A. Is available only for export
    B. Provides reliable 24-hour electricity using volcanic heat resources
    C. Has no installation cost
    D. Exists equally in every country
    ☑ **B** — The Philippines is rich in geothermal resources due to its volcanic activity.
23. Why is climate adaptation still necessary even if emissions are reduced?
    A. Because mitigation does not work
    B. Because some climate change is already unavoidable due to past emissions
    C. Because laws require it
    D. There is no specific reason
    ☑ **B** — Existing greenhouse gases will continue affecting the climate.
24. Renewable energy benefits the economy by:
    A. Increasing dependence on imported fuels
    B. Creating jobs, reducing fuel imports, and stabilizing energy prices
    C. Slowing economic growth
    D. Requiring permanent government subsidies
    ☑ **B** — Renewable energy supports both economic and environmental goals.
25. Which action contributes to BOTH climate mitigation and adaptation?
    A. Turning off lights only
    B. Planting native trees, building green roofs, and harvesting rainwater
    C. Driving less only
    D. Using fewer plastic bags only
    ☑ **B** — These actions reduce emissions while helping communities adapt.
26. The Philippines has renewable energy potential that is:
    A. Very limited
    B. Greater than the country's current electricity demand
    C. Enough only for rural communities
    D. Smaller than coal resources
    ☑ **B** — The country has abundant renewable energy resources.
27. Climate justice recognizes that:
    A. Every country contributes equally to climate change
    B. Countries with lower emissions often suffer the greatest climate impacts and deserve support
    C. Only developed countries should act
    D. Climate change affects everyone equally
    ☑ **B** — It focuses on fairness between those causing and experiencing climate impacts.
28. Why do individual daily choices matter in addressing climate change?
    A. Individual actions have no effect
    B. Many small actions together can reduce emissions and influence businesses and governments

C. They only matter in school activities
D. Governments are the only ones responsible
☑ **B** — Collective individual actions create significant change.

29. One of the biggest challenges to expanding renewable energy in the Philippines is:
A. Lack of sunlight and wind
B. Existing coal contracts, slow permit processes, and limited power grid capacity
C. Renewable energy being too inexpensive
D. Complete public opposition to renewables
☑ **B** — Infrastructure and policy barriers slow renewable energy growth.

30. Why is sustainability important for young people today?
A. It only matters for school examinations
B. Today's environmental decisions will shape future safety, livelihoods, and quality of life
C. It is only the responsibility of adults
D. It has no personal impact
☑ **B** — The future effects of climate change will be experienced by today's younger generations.